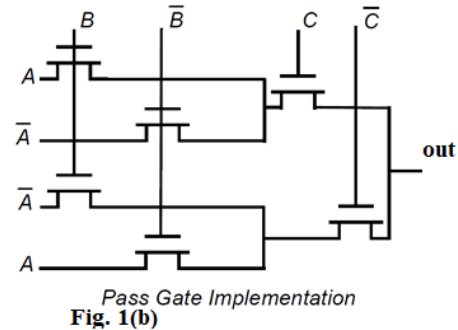
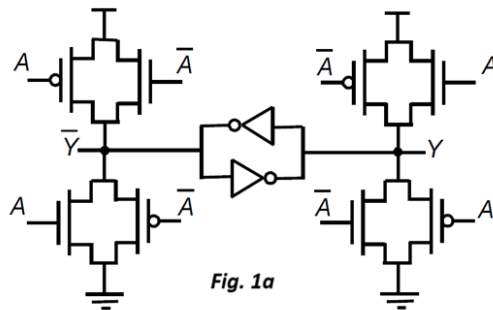


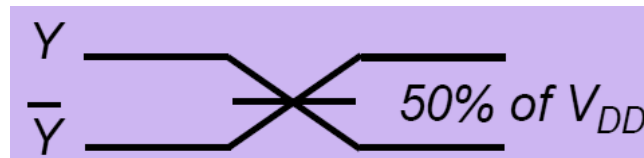
SECTION - A (55 marks total)

Q1. (a). Determine the Boolean Functions Y and $\bar{Y}$ implemented by the circuit shown in **Fig's 1(a) and 1(b)**.



Solution----fig1a)--- Differential Buffer, $Y = A$, $\bar{Y} = \bar{A}$

A	Y	$\bar{Y}$
0	0	1
1	1	0



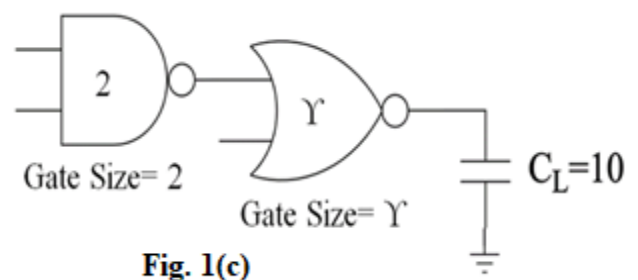
Solution—fig. 1(b)---

$$ABC + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

Q1. (b) Given that minimum sized standard CMOS inverter has $W_N=1$ and $W_P=2$, (take this gate has gate size=1)

Determine the value of Y for minimum path delay -of the 2-stage circuit shown in **Fig. 1(c)**.

Also, determine minimum path delay in delay units.



Solution: Q1b, Fig 1(c)

$$d_1 = g_{nand2}h_1 + p_{nand2} = \frac{4}{3} \frac{C_{in}(nor2)}{C_{in}(nand2)} + 2$$

$$d_2 = g_{nor2} h_2 + p_{nor2} = \frac{5}{3} \frac{C_L}{C_{in}(nor2)} + 2$$

$$d_1 = \frac{4}{3} \frac{5\gamma}{8} + 2 = \frac{5\gamma}{6} + 2$$

$$d_2 = \frac{5}{3} \frac{10}{5\gamma} + 2 = \frac{10}{3\gamma} + 2$$

$$D(\gamma) = d_1 + d_2 = \frac{5\gamma}{6} + \frac{10}{3\gamma} + 4$$

$$\frac{dD}{d\gamma} = \frac{5}{6} - \frac{10}{3\gamma^2}$$

$$\frac{5}{6} - \frac{10}{3\gamma^2} = 0$$

$$\gamma^2 = 4.$$

$$\hat{D} = D(2) = \frac{5 \cdot 2}{6} + \frac{10}{3 \cdot 2} + 4 = \frac{22}{3}$$

Q1 (c) Consider the circuits shown in Figure 1 (d) and Figure 1 (e). For the given input combinations, plot node voltages V_X and V_Y .

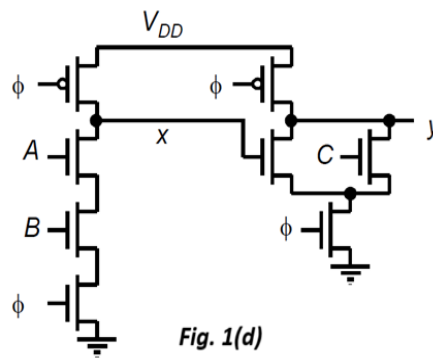


Fig. 1(d)

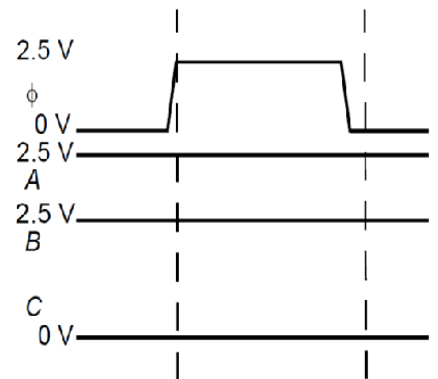
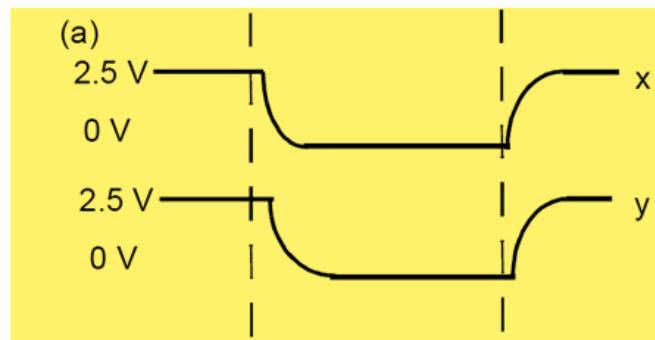


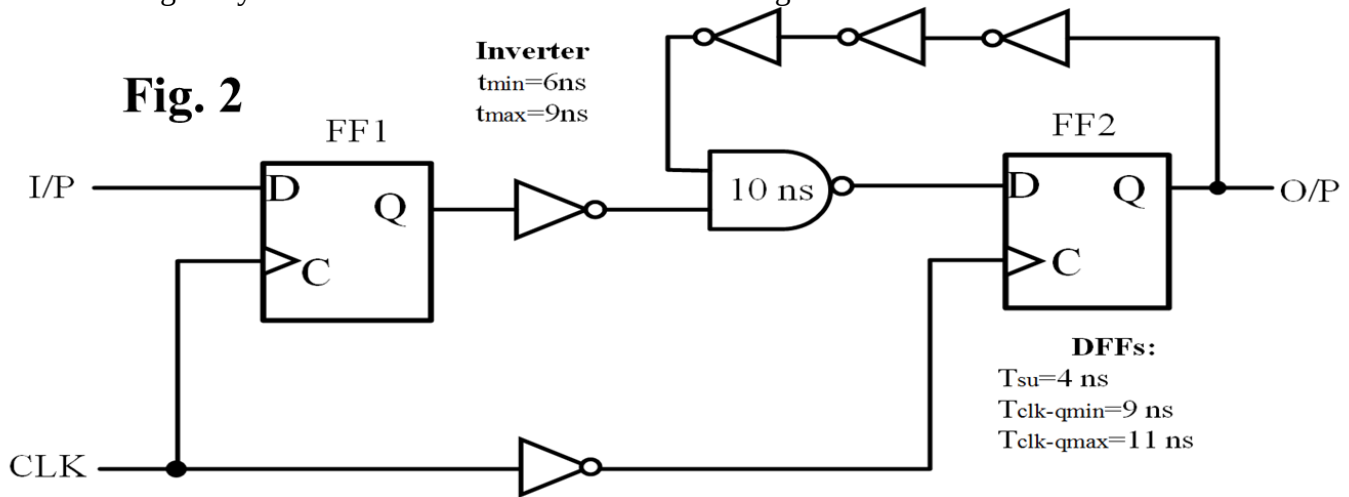
Fig. 1(e)

[19 marks]

Solution- Q1(c)---



Q2. Consider the circuit shown in **Fig. 2**, where the **flip-flop delays** and **gate delays** are annotated. Perform a detailed timing analysis of the circuit and answer the following:



Determine the maximum frequency of operation for the circuit in MHz. Show all steps and calculations.

- Find the hold time requirement** for the circuit. Provide a detailed explanation and calculations to support your answer.
- Assume a **clock skew of 9 ns** is introduced between Flip-Flop 1 (FF1) and Flip-Flop 2 (FF2). Analyze whether a hold time violation occurs under this condition. Clearly, calculate the hold time constraint and explain your reasoning.

Solution-

$$\frac{T_{clk}}{2} \geq 27ns + 10ns + 4ns \geq 41ns$$

$$T_{clk} \geq 82ns, f_{clk} \geq \underline{12.19 MHz}$$

$$\frac{T_{clk}}{2} \geq 41ns - 9ns \geq 32ns$$

$$\underline{T_{clk} = 64ns}, f_{clk} \geq \underline{15.62 MHz} \rightarrow \text{10M}$$

$$\begin{aligned} T_{hold} &< T_{clk-q_{min}} + T_{im(q_{min})} + 10ns \\ &< 9ns + 6ns + 10ns < \underline{25ns} \end{aligned} \quad \text{eq ①}$$

→ Possibility of hold time violation for T_{hold} ~~2.5ns~~.

→ But, if we analyze clk path then FF2 (P/K) becomes disabled (hold mode) after a worst case delay of $(INV) \approx 9ns$. $\rightarrow \text{4M}$

∴ Therefore, there is no hold time violation

3> An addition of extra 9ns of clk skew in path (FF1, FF2) will delay the triggering of second FF2 by 9ns. ∴ FF2 goes in hold mode after 18ns

∴ According to eq ①: There is no hold time violation. $\text{4M} \leftarrow$

Q3. Consider the design of a CMOS compound (OAI22) gate in a 22 nm technology node to implement the Boolean function appended below:

$$F = \bar{A}.\bar{B} + \bar{C}.\bar{D}$$

Given that the complimentary versions of inputs are not available, $(W/L)_n=20$, and $(W/L)_p=3(W/L)_n$.

- Sketch its transistor-level schematic.
- Sketch the corresponding stick diagram and properly label the source, polysilicon, & drain region of each transistor.
- Estimate the cell area from the stick diagram.
- Calculate the parasitic junction capacitance at the output node. (Here, please ignore the overlap capacitance in the channel region. The area junction capacitance is $C_j = 3 \times 10^{-4} \text{ pF}/\mu\text{m}^2$, and the side wall junction capacitance is $C_{jsw} = 2 \times 10^{-4} \text{ pF}/\mu\text{m}$). Consider the length of the source and drain regions as per the λ -based design rules tabulated below.
- Draw the cross-sectional diagram of the transistors connected in the n-diffusion region.

[19 marks]

λ -based design rules:

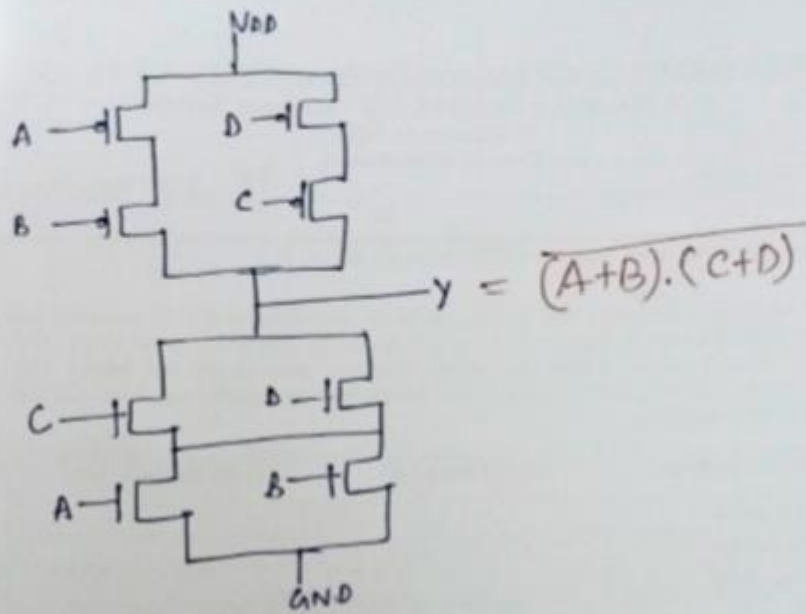
Active contact size	2λ
Minimum active contact to metal edge spacing	λ
Minimum active contact to poly edge spacing	3λ
Minimum active contact to active edge spacing	λ

Solution-----Q3

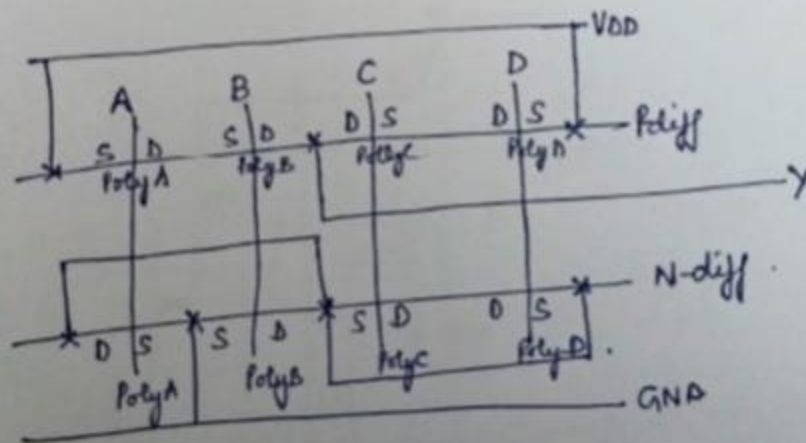
Solution

Q1

a)



b)



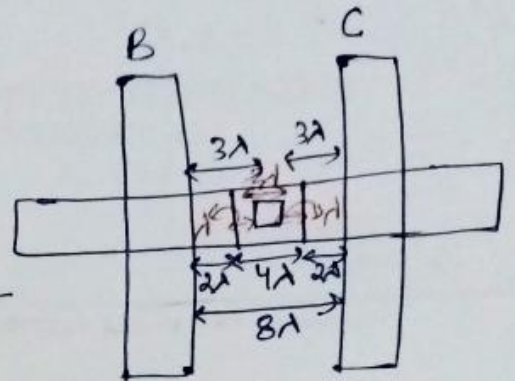
c) $L = 2\lambda = 22 \text{ nm}$

$\lambda = 11 \text{ nm}$

$5 \times 6 \text{ tracks} \Rightarrow 1 \text{ track} = 8\lambda$

$= 5 \times 8\lambda \times 6 \times 8\lambda$

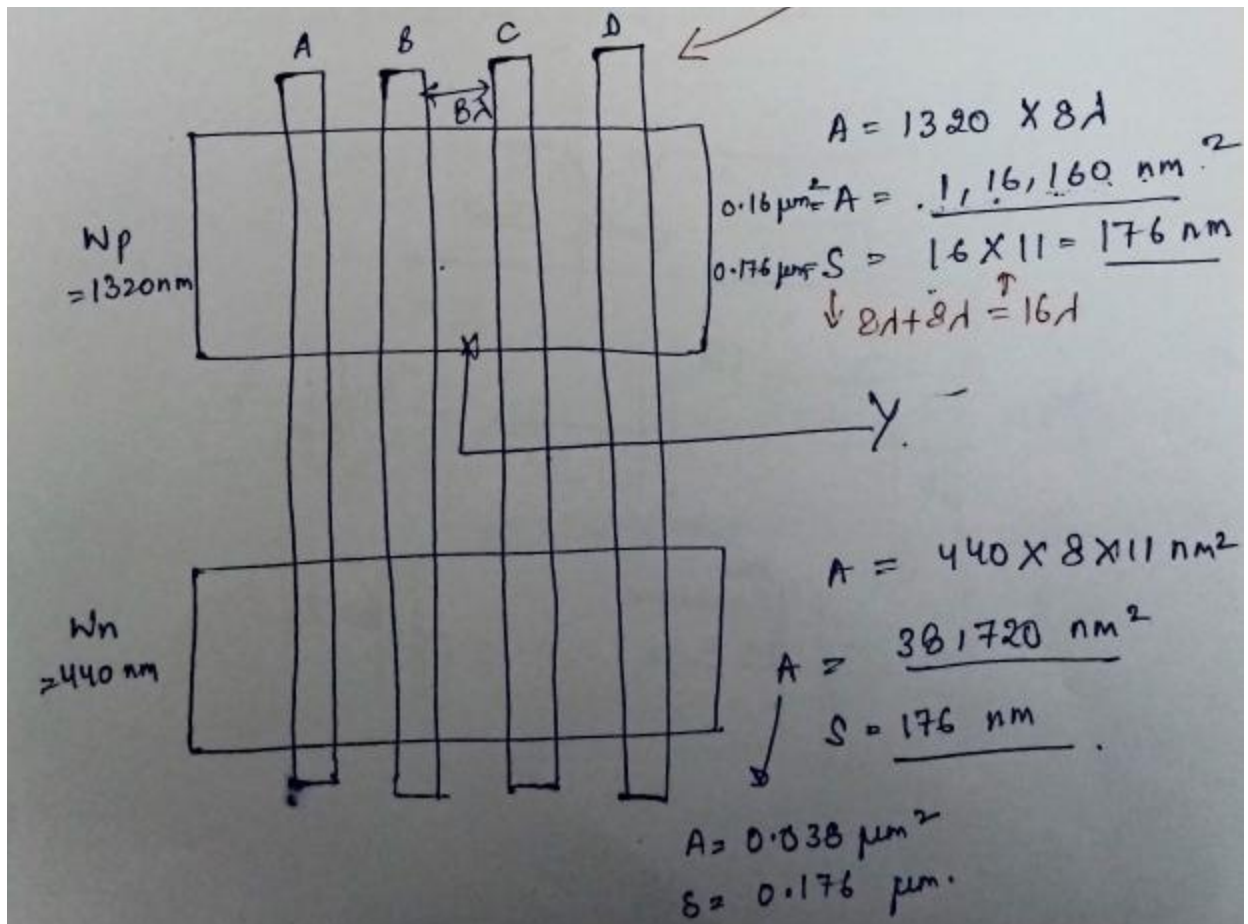
Area of cell $= 1920 \lambda^2 = 1920 \times 11^2 \text{ nm}^2$
 $= 232320 \text{ nm}^2$
 $= 0.232320 \mu\text{m}^2$



d) $\left(\frac{W}{L}\right)_n = 20$

$W_n = 20 \times 22 = 440 \text{ nm}$

$W_p = 3 \times 440 \text{ nm} = 1320 \text{ nm}$



$$\begin{aligned}
 C_0 &= (C_{\text{pull up}})_p + (C_{\text{pull down}})_p \\
 &= (A \cdot C_j + S \cdot C_{jsw})_{\text{pull up}} + (A \cdot C_j + S \cdot C_{jsw})_{\text{pull d}} \\
 &= 0.16 \mu\text{m}^2 \times \frac{3 \times 10^{-4} \text{ pf}}{\mu\text{m}^2} + 0.176 \mu\text{m} \times \frac{2 \times 10^{-4} \text{ pf}}{\mu\text{m}}
 \end{aligned}$$

$$\begin{aligned}
 C_0 &= (C_{j\text{up}})_p + (C_{j\text{down}})_p \\
 &= (A \cdot C_j + S \cdot C_{j\text{sw}})_{\text{pull up}} + (A \cdot C_j + S \cdot C_{j\text{sw}})_{\text{pull down}} \\
 &= 0.16 \mu\text{m}^2 \times 3 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}^2} + 0.176 \mu\text{m} \times 2 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}}
 \end{aligned}$$

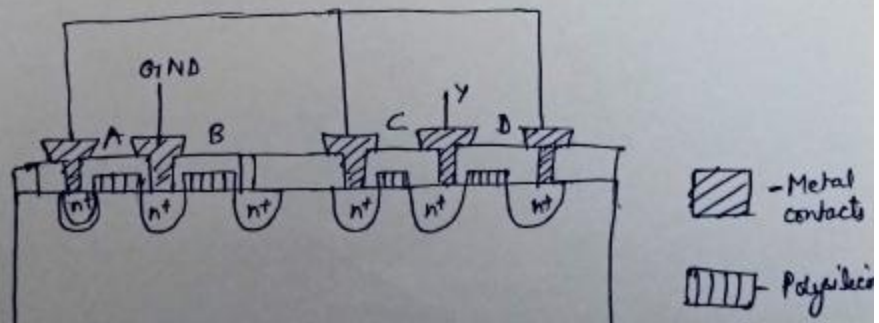
+

$$0.038 \mu\text{m}^2 \times 3 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}^2} + 0.176 \mu\text{m} \times 2 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}}$$

$$\begin{aligned}
 &= 0.000048 \text{ pF} + 0.0000352 \text{ pF} \\
 &\quad + 0.0000114 \text{ pF} + 0.0000352 \text{ pF}
 \end{aligned}$$

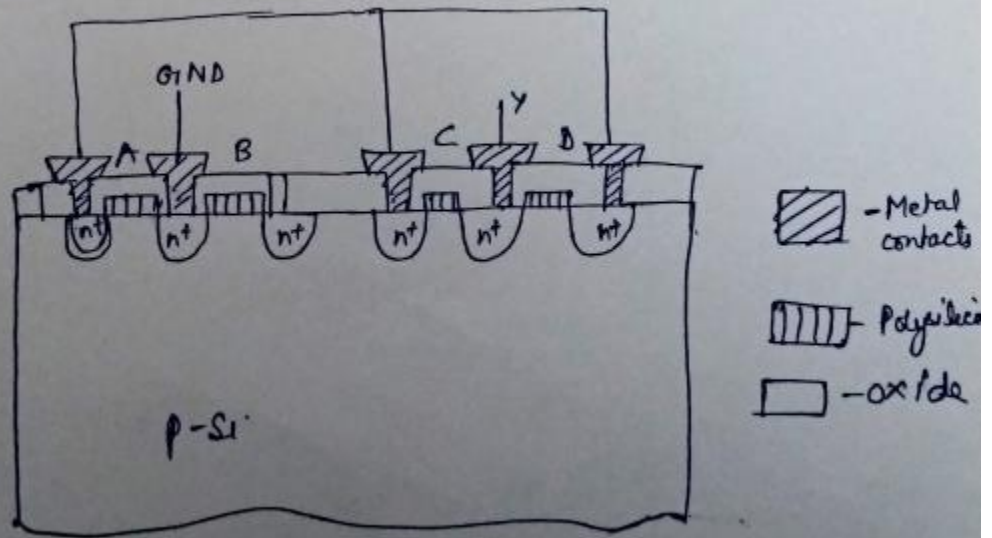
$$\underline{C_0 = 0.0001298 \text{ pF}} = \underline{0.1298 \text{ fF}} \quad \checkmark$$

e)



$$C_0 = 0.0001298 \text{ pF} = 0.1298 \text{ fF}$$

e)



-----Section A ends-----

Write the section name clearly on top of each sheet.

Answer the sub-parts of a question at one place and in the order in which they appear.

SECTION - B (Total 50 marks)

Common data: Use the following common data if not mentioned specifically in the question

For 0.25 μ m Technology node -- $L_{min}=250$ nm, pico (p) = 10^{-12} , femto (f) = 10^{-15}

Neglect body effect

Take $V_{DD}=2.5$ V, $V_{TON}=0.4$ V, $V_{TOP} = -0.4$ V, $\lambda_n=\lambda_p= 0.1$ V $^{-1}$, $\mu_n C_{ox}=90$ uA/ V 2 , $\mu_p C_{ox}= 30$ uA/ V 2

Q4. For the 2-stage op amp shown in **Fig. 4**, assume a power budget of 6 mW, a required output swing of 2.5 V, and $L_{eff} = 0.5$ μ m for all devices.

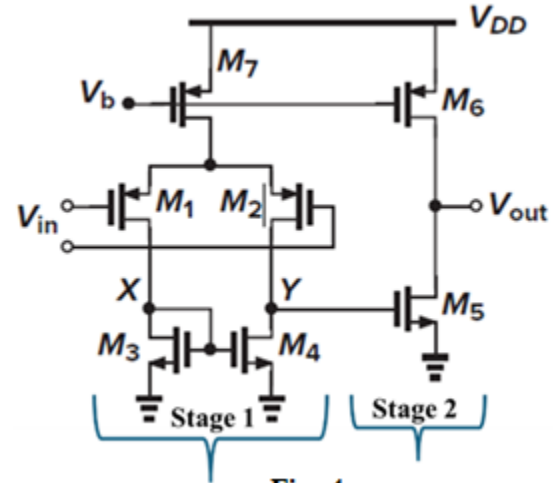
Given, $V_{DD} = 3$ V

for nMOS: $\lambda = 0.1$ V $^{-1}$, $V_{Tn} = 0.7$ V, $\mu_n C_{ox} = 134$ μ A/V 2 .

For pMOS: $\lambda = 0.1$ V $^{-1}$, $V_{Tp} = -0.7$ V, $\mu_p C_{ox} = 40$ μ A/V 2 .

- Allocating a current of 1 mA to the output stage and equal overdrive voltages to M_5 and M_6 , determine $(W/L)_5$ and $(W/L)_6$.
- Calculate the small-signal gain of the output stage (V_{out}/V_y).
- Determine the aspect ratio (W/L) of M_3 and M_4 such that $V_{GS3} = V_{GS5}$.
- Calculate the aspect ratios of M_1 and M_2 such that the overall voltage gain of the op amp is equal to 500.

[20 marks]



Solution---4---

Given: $V_{DD} = 3V$, $P = 6mW$,

$$V_{DD} - 2V_{OV} = 2.5V \Rightarrow V_{OV} = 0.25V$$

$$V_{OV,5} = V_{OV,6} = 0.25V$$

a) $1mA = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_5 V_{OV}^2 \rightarrow \text{for } M_5$

$$= \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L}\right)_6 V_{OV}^2 \rightarrow \text{for } M_6$$

$$\left(\frac{W}{L}\right)_5 = \frac{2 \times 1mA}{(\mu_n C_{ox}) V_{OV}^2} = \frac{2 \times 10^{-3}}{134 \times 10^{-6} \times 0.25^2} \approx \underline{239}$$

$$\left(\frac{W}{L}\right)_6 = \frac{2mA}{\mu_p C_{ox} V_{OV}^2} = \frac{2 \times 10^{-3}}{40 \times 10^{-6} \times 0.25^2} \approx \underline{800} \quad [4M]$$

b) 1st stage gain $A_{v1} = g_{m1} (r_{o2} || r_{o4}) =$

$$\text{2nd stage gain } A_{v2} = g_{m5} (r_{o5} || r_{o6}) = 8 \times 10^{-3} \times (3.33 \times 10^3) \\ = 26.64 \text{ V/V}$$

$$r_{o5} = \frac{1}{\lambda_n I_{D5}} = \frac{1}{0.1 \times 1mA} = 10 \text{ K}\Omega \quad [4M]$$

$$r_{o6} = \frac{1}{0.2 \times 1mA} = 5 \text{ K}\Omega$$

$$g_{m5} = \frac{2I_D}{V_{OV}} = \frac{2mA}{0.25V} = 8 \text{ mS}$$

1mA through M_7 , will be divided equally (0.5mA) in 2 branches.

$$r_{o2} = 20 \text{ K}\Omega, r_{o6} = 10 \text{ K}\Omega, g_{m1} = \frac{2I_D}{V_{OV}} ?$$

$$c) \quad V_{bs5} = V_{bs3} \Rightarrow V_{ov3} = 0.25V$$

$$(W/L)_{3,4} = \frac{2 \times 0.5mA}{134 \times 10^{-6} \times 0.25^2} \approx 120 \quad [4M]$$

$$d) \quad g_{m1} = \frac{2 I_{D1}}{V_{ov1}} = 2 \times$$

$$g_{m1} = \sqrt{2 \mu_p C_{ox} (W/L)_1 I_{D1}} \quad [4M]$$

$$\Rightarrow (W/L)_1 = \frac{g_{m1}^2}{2 \mu_p C_{ox} I_{D1}} = \frac{2.82^2 \times 10^3}{2 \times 40 \times 0.5} = 199$$

$$A_{v1} \cdot A_{v2} = 500 \Rightarrow A_{v1} = \frac{500}{26.64} = 18.8$$

$$g_{m1} (R_{o2} || R_{o4}) = 18.8$$

$$\Rightarrow g_{m1} = \frac{18.8}{6.67} mS = 2.82 mS$$

Q5. (A) Consider the logic circuit shown in **Fig. 5**, with threshold voltage V_{TO} (NMOS/ driver) = 1 V, V_{TO} (PMOS/load) = - 3 V, and body effect coefficient (γ) = 0.

- i). Determine the logic function F.
- ii). Calculate (W/L)_p such that minimum output voltage (V_{OL}) does not exceed 0.4 V.

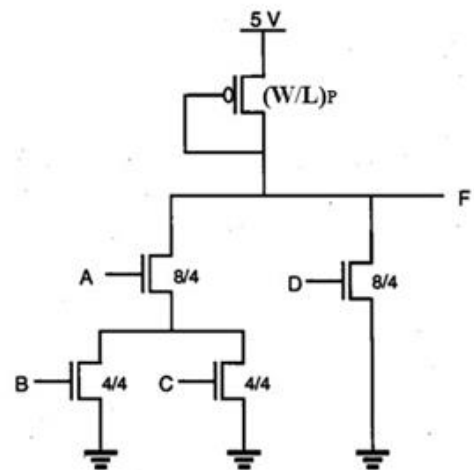


Fig. 5

Solution 5 (A)---

Solve 1) A) (i)

$$F = \overline{A(B+C)+D}$$

(ii) V_{OL} Maximum occurs when $A=1$, $D=0$ and either (but not both) $B=1$ or $C=1$. Under these conditions, the circuit may be simplified as a depletion load inverter with the equivalent driver W/L given below.

$$\left(\frac{W}{L}\right)_{eq} = \frac{1}{\left(\frac{4}{8}\right) + \left(\frac{4}{4}\right)} = \frac{1}{3/2}$$

$$\boxed{\left(\frac{W}{L}\right)_{eq} = \frac{2}{3}}$$

When $V_{OL} = 0.4$, $V_{in} = 5$, the load is saturated and the driver is in linear region.

$$\frac{K'}{2} \left(\frac{W}{L}\right)_{load} (V_{DS,load} - V_{T,load})^2 = K' \left(\frac{2}{3}\right) \left[(V_{in} - V_{T,driver}) V_{out} - \frac{V_{out}^2}{2} \right]$$

$$\Rightarrow \frac{K'}{2} \left(\frac{W}{L}\right)_{load} (0+3)^2 = K' \left(\frac{2}{3}\right) \left[(5-1) \times 0.4 - \frac{1}{2} \times 0.4^2 \right]$$

$$\boxed{\left(\frac{W}{L}\right)_{load} = 0.225}$$

$$\therefore \left(\frac{W}{L}\right)_{load} \leq 0.225 \text{ for } V_{OL} \leq 0.4$$

B) Consider a CMOS inverter with the following parameters:

Given: $V_{DD} = 1.2 \text{ V}$, The channel length of both transistors is $L_n = L_p = 60 \text{ nm}$.

nMOS $V_{T0,n} = 0.48 \text{ V}$, $\mu_n C_{ox} = 102 \mu\text{A/V}^2$, $E_{c,n} L_n = 0.4 \text{ V}$, $(W/L)_n = 10$

pMOS $V_{T0,p} = -0.46 \text{ V}$, $\mu_p C_{ox} = 51.6 \mu\text{A/V}^2$, $E_{c,p} L_p = 1.8 \text{ V}$, $(W/L)_p = 19$

i) . Determine the (W_n / W_p) ratio so that the switching (inversion) threshold voltage of the circuit is

$$V_{th} = 0.5 \text{ V}.$$

ii) . The CMOS fabrication process used to manufacture this inverter allows a variation of the $V_{T0,n}$ value by $\pm 15\%$ around its nominal value, and a variation of the $V_{T0,p}$ value by $\pm 20\%$ around its nominal value.

Assuming that all other parameters (such as μ_n , μ_p , C_{ox} , W_n , W_p) always retain their nominal values, find the upper and lower limits of the **switching threshold voltage (V_{th})** of this circuit.

[18 marks]

Solution 5B----

~~Solve (A)~~

(i) $V_{DS, sat, n} = \frac{(V_{th} - V_{T0, n}) \cdot E_{cn} \cdot L_n}{(V_{th} - V_{T0, n}) + E_{cn} \cdot L_n} = \frac{(0.5 - 0.48) \cdot 0.4}{(0.5 - 0.48) + 0.4} = 19 \text{ mV}$

$V_{SD, sat, p} = \frac{(V_{DD} - V_{th} - |V_{T0, p}|) \cdot E_{cp} \cdot L_p}{(V_{DD} - V_{th} - |V_{T0, p}|) + E_{cp} \cdot L_p}$

$= \frac{(1.2 - 0.5 - 0.46) \cdot 1.8}{(1.2 - 0.5 - 0.46) + 1.8} = 212 \text{ mV}$

$\Rightarrow \therefore K_R = \frac{K_n}{K_p} = \left(\frac{V_{DD} - |V_{T0, p}| - V_{SD, sat, p} - V_{th}}{V_{th} - V_{T0, n} - V_{DS, sat, n}} \right) \left(\frac{E_{cp}}{E_{cn}} \right)$

$K_R = \left(\frac{1.2 - 0.46 - 0.212 - 0.5}{0.5 - 0.48 - 0.019} \right) \left(\frac{1.8}{0.4} \right) = 12.6$

$\therefore K_R = \frac{K_n}{K_p} = \frac{\mu_n C_{ox} \left(\frac{W}{L} \right)_n}{\mu_p C_{ox} \left(\frac{W}{L} \right)_p} = \frac{102}{51.6} \frac{W_n}{W_p} = 12.6$

$\Rightarrow \boxed{\frac{W_n}{W_p} = 64}$

(ii) $V_{T0, n} = 0.552 (15\%)$

$V_{T0, p} = -0.552 (20\%)$

$K_R = 1.04 = \frac{\mu_n C_{ox} \left(\frac{W}{L} \right)_n}{\mu_p C_{ox} \left(\frac{W}{L} \right)_p}$

$V_{th} = \frac{V_{T0, n} + \sqrt{\frac{1}{K_R}} (V_{DD} + V_{T0, p})}{\left(1 + \sqrt{\frac{1}{K_R}} \right)} = \frac{0.552 + \sqrt{\frac{1}{1.04}} (1.2 - 0.552)}{\left(1 + \sqrt{\frac{1}{1.04}} \right)}$

$\boxed{V_{th} = 0.600 \text{ V}}$

Case b) $V_{T0,n} = 0.408V (-15\%)$ $K_R = 1.04$
 $V_{T0,p} = -0.552V (20\%)$

$$V_{th} = \frac{0.408 + \sqrt{\frac{1}{1.04}} (1.2 - 0.552)}{\left(1 + \sqrt{\frac{1}{1.04}}\right)} = \boxed{0.527V}$$

Case c) $V_{T0,n} = 0.552V (15\%)$ $K_R = 1.04$
 $V_{T0,p} = -0.368V (-20\%)$

$$V_{th} = \frac{0.552 + \sqrt{\frac{1}{1.04}} (1.2 - 0.368)}{1 + \sqrt{\frac{1}{1.04}}}$$

$$\boxed{V_{th} = 0.691V}$$

Case d) $V_{T0,n} = 0.408V (-15\%)$ $K_R = 1.04$
 $V_{T0,p} = -0.368V (-20\%)$

$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{K_R}} (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{1}{K_R}}\right)}$$

$$= \frac{0.408 + \sqrt{\frac{1}{1.04}} (1.2 - 0.368)}{\left(1 + \sqrt{\frac{1}{1.04}}\right)}$$

$$\boxed{V_{th} = 0.618V}$$

So upper limit $0.691V$
 lower limit $0.527V$

Q6. Consider shown in **Fig. 6**. Take $V_{ov} = 0.2V$. Assume capacitive load at node V1 is **14 pF**, and capacitive load at node Vout is **2pF**

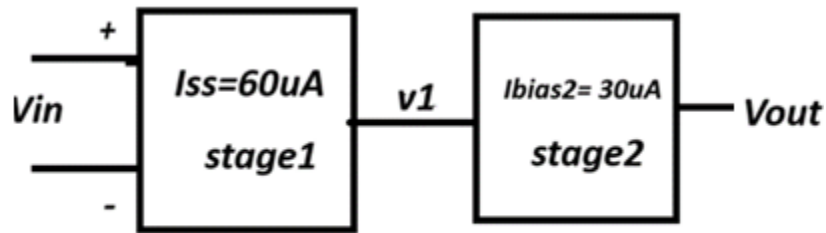


Fig. 6

a) Sketch and label the schematic of **Fig. 6** for the following cases—

- i). **Case-1**---- stage1 is current mirror load differential amplifier, and stage2 is pmos input cascode amplifier
- ii). **Case-2**---- stage1 is current mirror load differential amplifier with cascode load, and stage2 is pmos input common source amplifier

Intuitively---

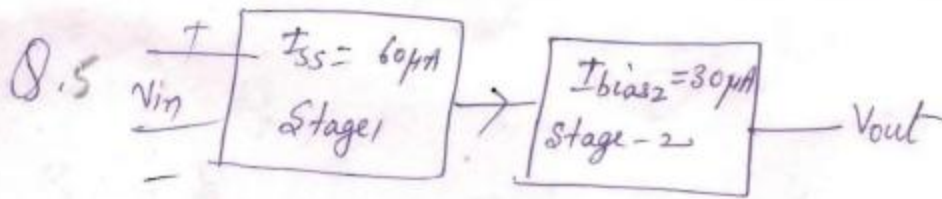
b) Determine low frequency small signal voltage gain $A_v = v_{out} / V_{in}$ for both **case-1**, and **case-2**

c) Determine the pole frequencies at node $v1$, and node V_{out} for both **case-1**, and **case-2**

d) Sketch and label Bode magnitude plot for both **case-1**, and **case-2**

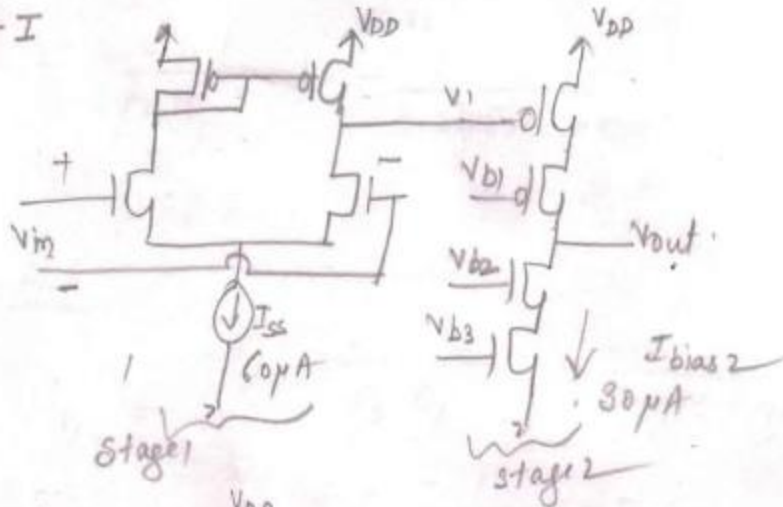
[17 marks]

Solution Q6----



$$V_{OV} = 0.2 V$$

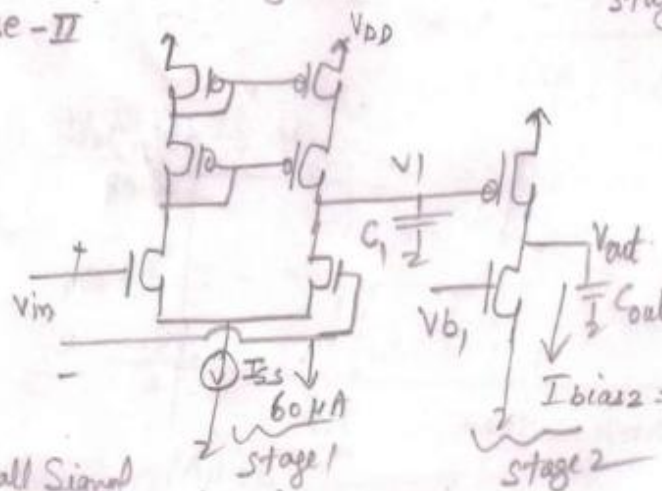
(a) Case - I



Case - II



Case-II



$$\lambda_n = \lambda_p$$

$$V_{ov_n} = V_{ov_p}$$

$$I_n = I_p$$

$$\Rightarrow g_{m_n} = g_{m_p} = \frac{2I}{V_{ov}} = \frac{2 \times 30 \mu A}{0.15 V} = 400 \mu A/V$$

$$\Rightarrow r_{o_n} = r_{o_p} = \frac{1}{\lambda I} = \frac{1}{0.005 \times 30 \mu A} = 666 K\Omega = 0.66 M\Omega$$

Small Signal
(b) Voltage Gain (L.F)

Case-1) $A_v = \frac{V_{out}}{V_{in}} = \text{Voltage Gain of Stage 1} \times \text{Voltage Gain of Stage 2}$

$$A_v = + g_{m_n} (r_{o_n} \parallel r_{o_p}) \times \left[- g_{m_p} (r_{o_p} \parallel r_{o_n}) \right]$$

$$\approx - g_{m_n}^2 \frac{r_{o_n}^2}{4} = - \frac{17.447 \times 10^{-6} \times 666^2}{4} \Rightarrow |A_v| = 19240 \approx 84.6 dB$$

Case-2

$$A_v = \frac{V_{out}}{V_{in}} = + g_{m_n} (r_{o_n} \parallel g_{m_p} r_{o_p} r_{o_n}) \times \left[- g_{m_p} (r_{o_p} \parallel r_{o_n}) \right]$$

$$\approx - g_{m_n}^2 \frac{r_{o_n}^2}{2} = - \frac{49.40 \times 10^{-6} \times 666^2}{2} \Rightarrow |A_v| = 10719 \approx 80.6 dB$$

pole frequencies at node v_1 & node V_{out} .

Case-1) ω_{p,v_1} , $\omega_{p,V_{out}}$

$$\omega_{p,v_1} = \frac{1}{\left(\frac{C_1}{2}\right) C_1} = \frac{1}{(0.139 \times 10^{-9}) \times 139 \times 10^{-12}} = 4.62 \times 10^6 \text{ rad/sec}$$

$$\omega_{p,V_{out}} = \frac{1}{\left(\frac{C_{out}}{2}\right) C_{out}} = \frac{1}{(0.2178 \times 10^{-9}) \times 2 \times 10^{-12}} = 50.4 \times 10^6 \text{ rad/sec}$$

UGB =

(d) Bode Plot

$$\omega_{p1} = \frac{1}{r_o C_1} = \frac{1}{0.66M \times 14p} = 108 K$$

Case 2 $C_{v1} =$

$$\omega_{p, \text{out}} = \frac{1}{\left(\frac{r_o}{2}\right) C_{out}} = \frac{1}{0.33M \times 2p} = 1.515 M$$

UGB =

